MSc Research Proposal – FAMB/FFCLRP-USP

Coherent and Dissipative Transport in Simple Open Quantum Networks

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Abstract

This project aims to study transport in small open quantum networks described as graphs. The main idea is to compare how different graph configurations behave when coherent hopping competes with local dephasing and capture into a sink. The goal is to understand, within a simple and controlled model, when coherence helps transport, when it hinders it, and how different network structures change that balance. The project will use a tight-binding Hamiltonian in the single-excitation sector and a Lindblad master equation as the starting framework. The central comparison will be carried out across a small set of graph families, using sink efficiency as the main observable and population dynamics plus coherence as secondary diagnostics. The emphasis is on physical clarity, systematic comparison between graph structures, and reproducible numerical results.

Keywords: open quantum systems; quantum transport; graph models; decoherence; Lindblad dynamics.

1. Introduction

Transport in quantum systems can already become nontrivial in very small models. If one prepares a single excitation in a network of coupled sites, its evolution depends both on the coherent hopping between sites and on the dissipative action of the environment. Even in a small graph, these ingredients can generate very different behaviors: fast transfer, destructive interference, trapping in inefficient pathways, or suppression of transport by strong decoherence [1–4].

The present proposal studies this problem in a deliberately simple way. The network will be treated as a graph with a small number of vertices. Different graph configurations — such as chains, rings, and complete graphs — will be compared under the same basic open-system dynamics. In this way, the project is not centered on one very specific geometry, but on how geometry itself shapes the competition between coherent and dissipative transport.

This is a good MSc problem because it is simple enough to be fully understood, but rich enough to produce nontrivial physics. It also allows a gradual progression: first understand the minimal model well, then compare different graph structures, and only later consider more elaborate extensions. The proposal is therefore centered on the smallest model that can already answer a real physics question: how topology and dephasing compete to determine transport toward a target channel in an open quantum network [5, 6].

2. Research question

The central question of the project is:

How do graph topology, coherent hopping, and local dephasing determine transport to a sink in simple open quantum networks?

More specifically, the project asks:

- how transport changes from one graph configuration to another;
- whether some graph structures are naturally more robust against decoherence than others;
- whether intermediate dephasing can improve transport in some cases.

3. Objectives

3.1. General objective

To study coherent and dissipative transport in simple open quantum networks by comparing different graph configurations within a common minimal model.

3.2. Specific objectives

1. Implement graph-based models with two to five sites.
2. Compare at least three graph families: chain, ring, and complete graph.
3. Describe coherent hopping, local dephasing, and sink capture in a common open-system framework.
4. Use sink efficiency as the main observable for comparing graph configurations.
5. Use population dynamics and coherence as secondary observables for interpreting the transport mechanism in each graph.
6. Determine how transport changes with the dephasing rate and with graph structure.
7. Identify parameter regimes in which dephasing suppresses or enhances transport.

4. Minimal model

4.1. Graph representation

The network will be described as a graph with N vertices. Each vertex represents a site that can host the excitation, and each edge represents a coherent hopping channel between two sites. In the single-excitation sector, the natural basis is $\{|i\rangle\}_{i=1}^{N}$, where $|i\rangle$ denotes the excitation localized at vertex i.

Different graph configurations correspond to different adjacency structures:

- in a chain, only nearest neighbors are connected;
- in a ring, the chain is closed periodically;
- in a complete graph, every site is connected to every other site.

This graph viewpoint is central to the project. The aim is not only to solve one transport model, but to compare how transport changes when the underlying graph changes.

4.2. Coherent dynamics

The coherent part of the model is given by the tight-binding Hamiltonian

$$H = \sum_{i=1}^{N} \epsilon_i |i\rangle\langle i| + \sum_{i \neq j} J_{ij} |i\rangle\langle j|. \tag{1}$$

Here, ϵ_i are site energies and J_{ij} are hopping amplitudes. The matrix J_{ij} is directly determined by the graph structure. Therefore, by changing the graph, one changes the coherent transport pathways available to the excitation.

4.3. Open-system dynamics

The dissipative evolution will be modeled by the Lindblad master equation

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right), \tag{2}$$

with $\hbar = 1$.

This is the simplest standard framework for combining coherent dynamics and dissipation in a physically controlled way [5, 6]. It is sufficient for a first study of the coherent-to-dissipative crossover in small networks.

4.4. Dissipative channels

The initial project will use two main dissipative mechanisms.

Local dephasing. At each site,

$$L_i^{(\phi)} = \sqrt{\gamma_{\phi,i}}\, |i\rangle\langle i|. \tag{3}$$

This channel preserves population but suppresses coherence between different sites.

Useful sink capture. Successful transport will be represented by an absorbing sink state $|s\rangle$, fed by a chosen target site $|t\rangle$:

$$L^{(\text{sink})} = \sqrt{\kappa}\, |s\rangle\langle t|. \tag{4}$$

The sink is included because the project is not about arbitrary spreading on a graph, but about transport toward a target channel.

Parasitic loss. In the numerical implementation, a separate loss channel is also available:

$$L_i^{(\text{loss})} = \sqrt{\Gamma_i}\, |\ell\rangle\langle i|. \tag{5}$$

This distinction is useful because it separates successful transfer from irrelevant dissipation. However, in the first stage of the MSc project, the central emphasis will remain on sink transport and dephasing. The loss channel can be treated initially as a secondary comparison layer.

4.5. Main observables

The main observable of the project will be the sink efficiency at time T,

$$\eta(T) = \rho_{ss}(T), \tag{6}$$

which measures the probability that the excitation has reached the sink by time T.

This will be the main quantity used to compare different graph configurations.

Two secondary observables will be used to interpret why one graph transports better than another.

Population dynamics. The site populations

$$P_i(t) = \rho_{ii}(t) \tag{7}$$

show how the excitation moves across the graph before reaching the sink.

Coherence. The site-basis coherence will be monitored through

$$C_{\ell_1}(\rho) = \sum_{i \neq j} |\rho_{ij}|. \tag{8}$$

Together, sink efficiency, population dynamics, and coherence are sufficient for the core version of the project. Quantities such as purity and entropy remain available in the codebase, but they will be treated as optional diagnostics rather than central observables.

4.6. Regimes of interest

The project will compare three physically distinct regimes:

- **Coherent regime:** dephasing is weak compared with the coherent hopping scale, so interference strongly affects the transport pathways.
- **Intermediate regime:** dephasing competes with coherent hopping without completely destroying coherence. This is the regime in which transport enhancement may appear in some graph topologies [2, 7–9].
- **Strongly dissipative regime:** dephasing dominates the dynamics and tends to suppress coherent transport.

This division is useful because it turns the parameter scan into a clear physical comparison rather than a purely numerical sweep.

5. Methodology

5.1. Step 1: smallest graphs

The project will begin with the simplest nontrivial graphs, such as two-site and three-site systems. The goal is to understand in detail how coherent hopping, sink coupling, and dephasing interact in the most elementary settings.

5.2. Step 2: comparison between graph families

The next step will compare chains, rings, and complete graphs with up to five sites. The goal here is to isolate the role of graph structure. Instead of treating topology as a secondary detail, the project makes topology itself one of the main control parameters.

5.3. Step 3: parameter sweeps

For each graph family, the dephasing rate will be varied systematically. This will make it possible to identify:

- coherent-dominated regimes;
- intermediate regimes in which dephasing changes the transport pattern;
- strongly dissipative regimes in which transport is suppressed.

5.4. Step 4: optional extensions

Once the minimal model is fully understood, simple extensions can be added:

- static disorder in the site energies;
- explicit comparison between useful sink capture and parasitic loss;
- reduced variables for a compact regime map.

These will be treated as natural developments, not as mandatory starting points.

6. Experimental motivation from superconducting circuits

Although the proposed MSc work is theoretical and computational, it is useful to keep contact with experimentally relevant platforms. Superconducting circuits provide such a motivation without changing the minimal structure of the project. They offer programmable couplings, controlled dissipation, and direct access to small network geometries, which makes them a natural long-term target for comparison between simulation and experiment.

This motivation is already supported by the literature. Potočnik *et al.* used superconducting circuits to emulate light-harvesting transport models [10]. More recently, Zhang *et al.* reported steady quantum transport in a superconducting processor [11]. In parallel, the open-system description of superconducting qubits continues to be an active methodological theme [12]. For the present proposal, this literature will be used only as a short experimental motivation and as a guide for plausible parameter windows. The core MSc problem remains the study of small graph models themselves.

7. Why this project is scientifically useful

The project becomes more useful if it is treated as a systematic comparison of graph-based transport models rather than as a collection of isolated numerical examples. Its scientific value comes from four features:

1. it compares different graph structures within the same physical model;
2. it keeps the model simple enough to be fully understood;
3. it can identify when decoherence helps or hinders transport;
4. it can be implemented as a reproducible computational benchmark.

This gives the project a clear path toward a future article: a systematic study of coherent and dissipative transport across simple graph families, with emphasis on physical interpretation, regime classification, and reproducibility.

8. Infrastructure already available

The project does not start from scratch. The *Quantum-systems* repository already contains:

1. a Lindblad solver,
2. routines for sink efficiency, population dynamics, coherence, and other basic open-system observables,
3. automated tests,

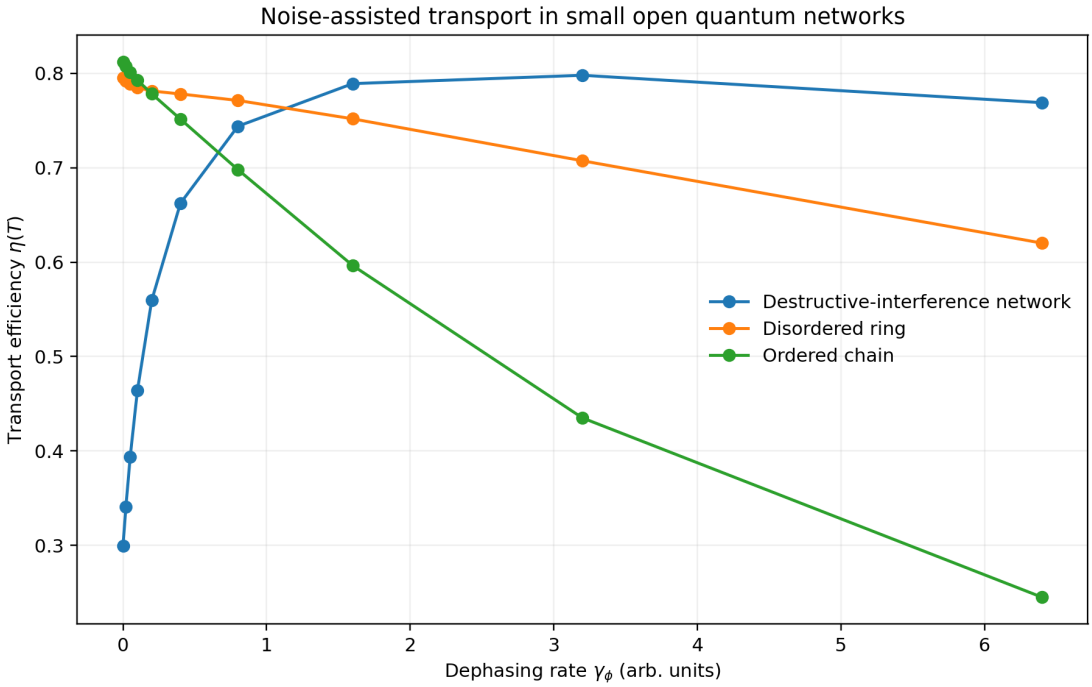

Figure 1: Main comparison figure: sink efficiency $\eta(T)$ as a function of the dephasing rate for three graph configurations. This is the principal observable of the project because it measures the fraction of excitations successfully captured by the sink. The curves already show that the same dissipative channel can enhance transport in one topology and suppress it in another.

4. a one-command transport laboratory that regenerates the main figures, metrics, and regime summaries for the proposal,
5. reproducible conventions for scripts, figures, metrics, and configuration files,
6. a LaTeX pipeline for documentation.

This means that the practical effort can be focused on the physics of the graph models, rather than on building numerical infrastructure from the ground up.

9. Preliminary results

As preparation for this proposal, a first graph-based transport module has already been implemented in the repository. The current baseline compares three simple network structures: a complete graph, a disordered ring, and an ordered chain.

The preliminary results already show that the graph structure matters strongly. In the complete graph, sink transport improves substantially at an intermediate dephasing rate. In the ring, the best performance remains near the coherent limit. In the chain, strong dephasing suppresses transport significantly.

This is exactly the type of behavior that motivates the project: the same dissipative mechanism can play very different roles depending on the underlying graph.

10. Timeline

The proposed 24-month timeline is:

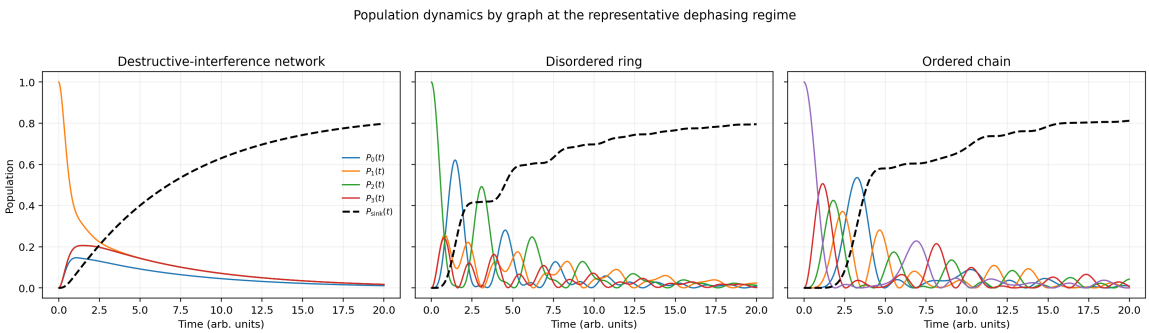

Figure 2: Secondary figure: population dynamics in the representative regime selected for each graph. The site populations $P_i(t)$ and the sink population $P_{\text{sink}}(t)$ show how the excitation propagates through the network before capture. This figure is useful for interpreting the transport pathways behind the sink-efficiency curves.

Period	Main activity
Months 1–3	Literature review and full consolidation of the minimal graph transport model.
Months 4–6	Detailed study of the smallest graphs and simplest topologies.
Months 7–9	Systematic comparison between chains, rings, and complete graphs under dephasing and sink capture.
Months 10–12	Extended parameter sweeps and interpretation of the coherent-to-dissipative crossover.
Months 13–15	Optional inclusion of disorder and explicit sink-versus-loss comparisons.
Months 16–18	Consolidation of the main results and first manuscript draft.
Months 19–21	Dissertation writing.
Months 22–24	Final revision, defense preparation, and reproducibility cleanup.

11. Expected outcomes

By the end of the project, the expected outputs are:

1. a clear physical account of coherent and dissipative transport in small graph-based quantum networks;
2. a systematic comparison between different graph configurations using sink efficiency as the main observable and population dynamics plus coherence as secondary diagnostics;
3. a reproducible computational benchmark with explicit coherent, intermediate, and strongly dissipative regimes;
4. a dissertation and, ideally, a paper based on the graph-comparison study.

12. Conclusion

This proposal is centered on a simple but physically meaningful problem: transport in small open quantum networks treated as graphs. Its strength is the combination of conceptual clarity, systematic topology comparison, and technical feasibility. That makes it a comfortable and well-structured theme for MSc supervision, while still leaving room for nontrivial physical results.

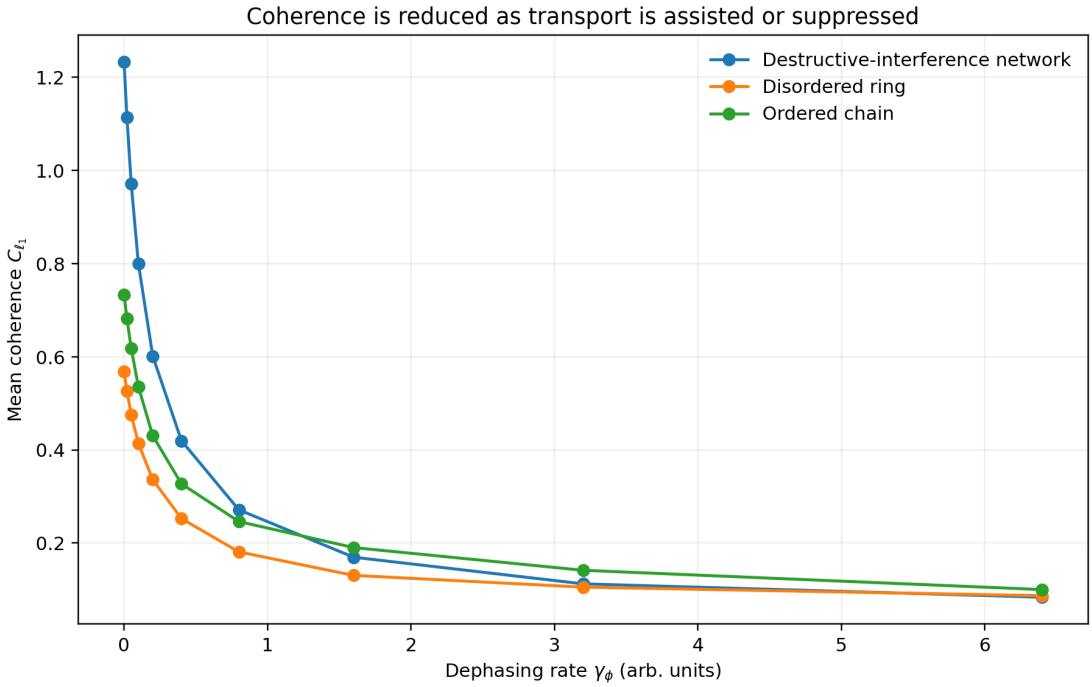

Figure 3: Secondary figure: mean coherence C_{ℓ_1} as a function of the dephasing rate for the same graph configurations. This diagnostic helps connect transport performance to the destruction or preservation of quantum coherence, without making coherence itself the principal performance metric.

References

[1] Masoud Mohseni, Patrick Rebentrost, Seth Lloyd, and Alán Aspuru-Guzik. Environment-assisted quantum walks in photosynthetic energy transfer. *The Journal of Chemical Physics*, 129(17):174106, 2008. doi: 10.1063/1.3002335.

[2] Martin B. Plenio and Susana F. Huelga. Dephasing-assisted transport: Quantum networks and biomolecules. *New Journal of Physics*, 10:113019, 2008. doi: 10.1088/1367-2630/10/11/113019.

[3] Patrick Rebentrost, Masoud Mohseni, Ivan Kassal, Seth Lloyd, and Alán Aspuru-Guzik. Environment-assisted quantum transport. *New Journal of Physics*, 11:033003, 2009. doi: 10.1088/1367-2630/11/3/033003.

[4] Filippo Caruso, Alex W. Chin, Animesh Datta, Susana F. Huelga, and Martin B. Plenio. Highly efficient energy excitation transfer in light-harvesting complexes: The fundamental role of noise-assisted transport. *The Journal of Chemical Physics*, 131(10):105106, 2009. doi: 10.1063/1.3223548.

[5] Heinz-Peter Breuer and Francesco Petruccione. *The Theory of Open Quantum Systems*. Oxford University Press, Oxford, 2007. doi: 10.1093/acprof:oso/9780199213900.001.0001.

[6] Ángel Rivas and Susana F. Huelga. *Open Quantum Systems: An Introduction*. Springer, Heidelberg, 2012. doi: 10.1007/978-3-642-23354-8.

[7] Alexandre R. Coates, Brendon W. Lovett, and Erik M. Gauger. Localisation determines the optimal noise rate for quantum transport. *New Journal of Physics*, 23(12):123014, 2021. doi: 10.1088/1367-2630/ac3b2c.

[8] Alexandre R. Coates, Brendon W. Lovett, and Erik M. Gauger. From goldilocks to twin peaks: multiple optimal regimes for quantum transport in disordered networks. *Physical Chemistry Chemical Physics*, 25(16):11134–11147, 2023. doi: 10.1039/D2CP04935J.

[9] Samuel L. Jacob, Laetitia P. Bettmann, Artur M. Lacerda, Krissia Zawadzki, Stephen R. Clark, John Goold, and Juan José Mendoza-Arenas. Dephasing-assisted transport in a tight-binding chain with a linear potential. *Frontiers in Physics*, 12:1474018, 2024. doi: 10.3389/fphy.2024.1474018.

[10] Anton Potočnik, Arno Bargerbos, Florian A. Y. N. Schröder, Saeed A. Khan, Michele C. Collodo, Simone Gasparinetti, Yves Salathé, Celestino Creatore, Christopher Eichler, Hakan E. Türeci, Alex W. Chin, and Andreas Wallraff. Studying light-harvesting models with superconducting circuits. *Nature Communications*, 9:904, 2018. doi: 10.1038/s41467-018-03312-x.

[11] Pengfei Zhang, Yu Gao, Xiansong Xu, Ning Wang, Hang Dong, Chu Guo, Jinfeng Deng, Xu Zhang, Jiachen Chen, Shibo Xu, Ke Wang, Yaozu Wu, Chuanyu Zhang, Feitong Jin, Xuhao Zhu, Aosai Zhang, Yiren Zou, Ziqi Tan, Zhengyi Cui, Zitian Zhu, Fanhao Shen, Tingting Li, Jiarun Zhong, Zehang Bao, Liangtian Zhao, Jie Hao, Hekang Li, Zhen Wang, Chao Song, Qiujiang Guo, H. Wang, and Dario Poletti. Emergence of steady quantum transport in a superconducting processor. *Nature Communications*, 15:10115, 2024. doi: 10.1038/s41467-024-54332-9.

[12] Hamid Reza Naeij. Open quantum system approaches to superconducting qubits. *Quantum Information Processing*, 24:220, 2025. doi: 10.1007/s11128-025-04838-y.